

Q-1 What is "Graph Based Clustering"?

Graph based clustering is a powerful technique that leverages graph theory to uncover hidden patterns and structures within data by representing data points as nodes, and the relationships between them as edges.

Explain Clustering with minimal spanning tree with example

Minimal spanning tree clustering is a technique that uses the concept of minimal spanning trees to divide data into clusters. Minimal spanning trees are trees that connect all points in a graph while minimizing the total edge weight.

While the original graph may be built using any high dimensional representation of the data, First level clusters are formed by partitioning an MST into sub trees by removing longest branches,

For example, below, suppose we start with 10 points and the 3-point with distance between the farthest pair of points among all edges. This produces two sub-trees. Thereafter, sub-trees that are too small in some meaningful way (e.g. only one-point or anti-cluster) are removed. If the remaining clusters themselves still contain more points, this process may be recursively repeated until all clusters contain only a satisfactory number of points.

Q.2 What is Model based Clustering?

Explain Expectation Maximization Algorithm with example.

Model based clustering: The aim of model based clustering is to maximize the likelihood of certain data belonging to a certain distribution. Expectation maximization (EM) is a popular algorithm for determining this maximum likelihood value. EM assumes that the data set generated by a mixture of distribution and estimates the means and co-variances of data for each component. EM algorithm proceeded in following steps:

E step: Provided the parameters for the components, calculate for each data point the probability that it belongs to changes its membership.

M step: Membership is revised to the nearest probability result from E step.

Steps repeat until stability.

Q. 3 Describe Density Based clustering (DBSCAN) with example

Density Based Clustering (DBSCAN) is a clustering algorithm that groups together data points that are close to each other based on a specified distance parameter (epsilon) and a minimum number of points (minPts) required to form a dense region.

Unlike K-Means, DBSCAN can identify clusters of arbitrary shapes and sizes.

For example, if $\epsilon = 0.5$ and minPts 5, DBSCAN would consider a point as core point if it has at least 5 neighbours within a radius of 0.5 units.